NATIONAL UNIVERSITY
of Computer & Emerging Sciences

DATABASE SYSTEMS

Topic: Lab 2
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Select Query Task:

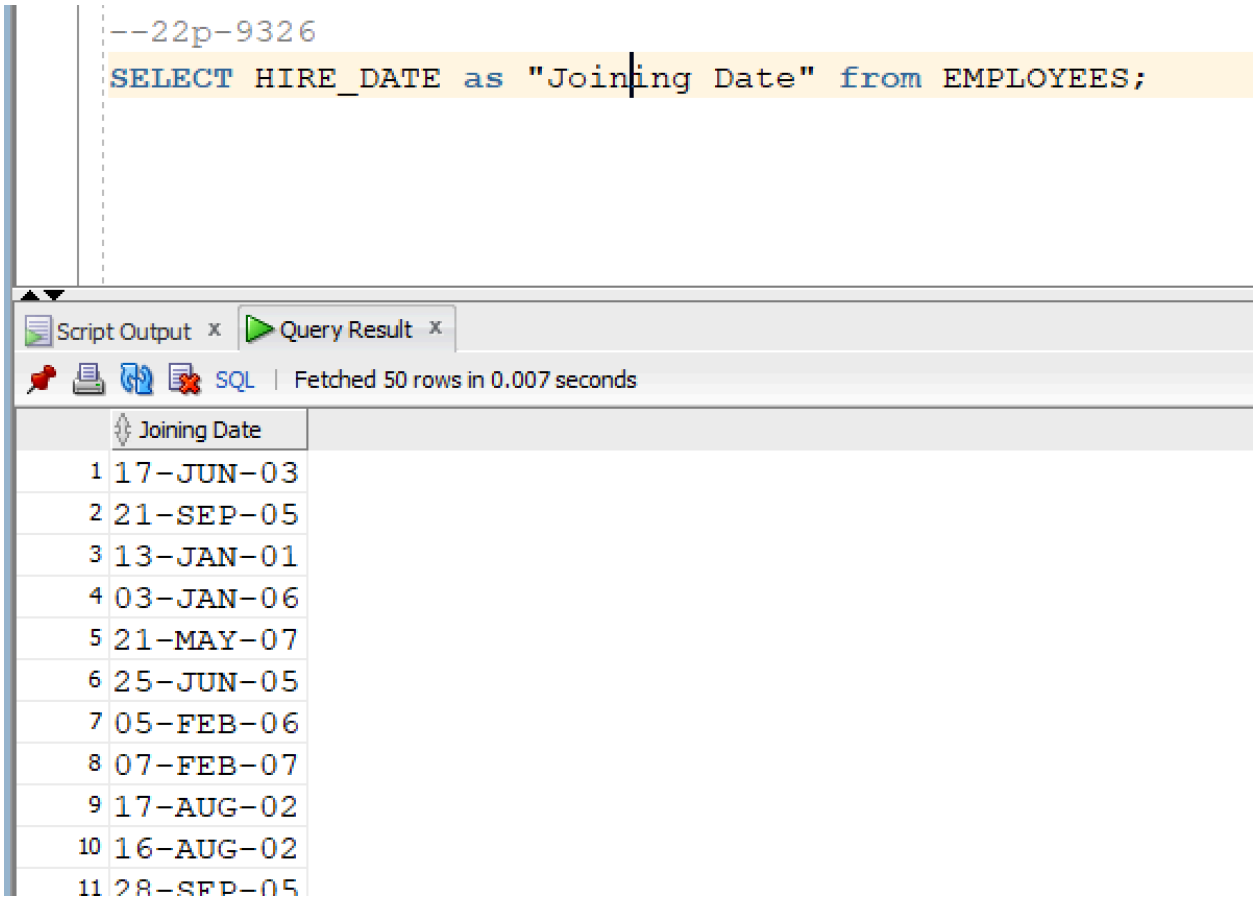
```
--22p-9326
SELECT employee_id, salary from EMPLOYEES;
```

Script Output | Query Result

SQL | Fetched 50 rows in 0.014 seconds

	EMPLOYEE_ID	SALARY
1	100	24000
2	101	17000
3	102	17000
4	103	9000
5	104	6000
6	105	4800
7	106	4800
8	107	4200
9	108	12008
10	109	9000
11	110	8200

Column Alias Query:



```
--22p-9326
SELECT HIRE_DATE as "Joining Date" from EMPLOYEES;
```

Script Output | Query Result

SQL | Fetched 50 rows in 0.007 seconds

	Joining Date
1	17-JUN-03
2	21-SEP-05
3	13-JAN-01
4	03-JAN-06
5	21-MAY-07
6	25-JUN-05
7	05-FEB-06
8	07-FEB-07
9	17-AUG-02
10	16-AUG-02
11	28-SEP-05

Concatenation Operator Query:

```
--22p-9326
SELECT FIRST_NAME || LAST_NAME as "Full Name" from EMPLOYEES;
```

Script Output | Query Result

SQL | Fetched 50 rows in 0.016 seconds

	Full Name
1	EllenAbel
2	SundarAnde
3	MozheAtkinson
4	DavidAustin
5	HermannBaer
6	ShelliBaida
7	AmitBanda
8	ElizabethBates
9	SarahBell
10	DavidBernstein

1) List all Employees having annual salary greater 20, 000 and lesser than 30,000.

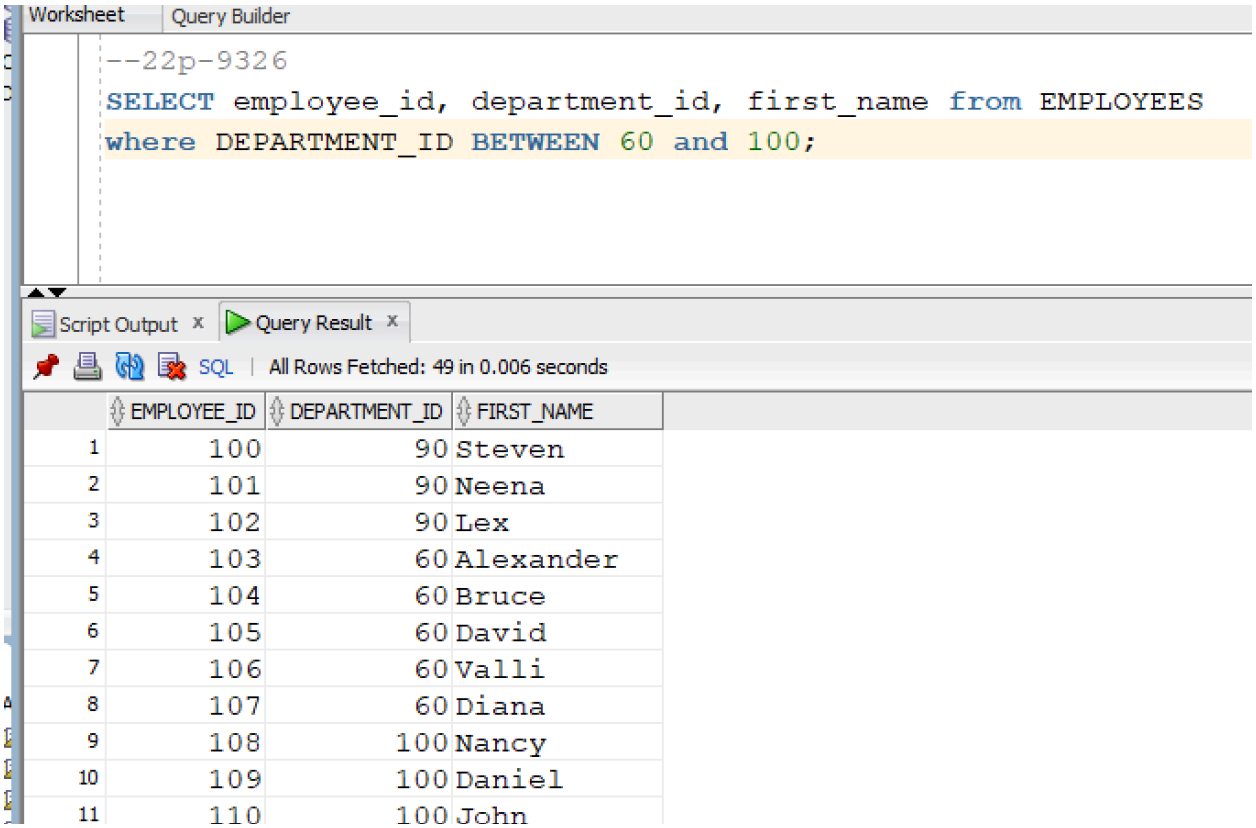
```
--22p-9326
SELECT salary from EMPLOYEES where SALARY > 20000 and
salary < 30000;
```

Script Output | Query Result

SQL | All Rows Fetched: 1 in 0.005 seconds

	SALARY
1	24000

2) List employee_id and first_name of employees from department # 60 to department #100.



Worksheet | Query Builder

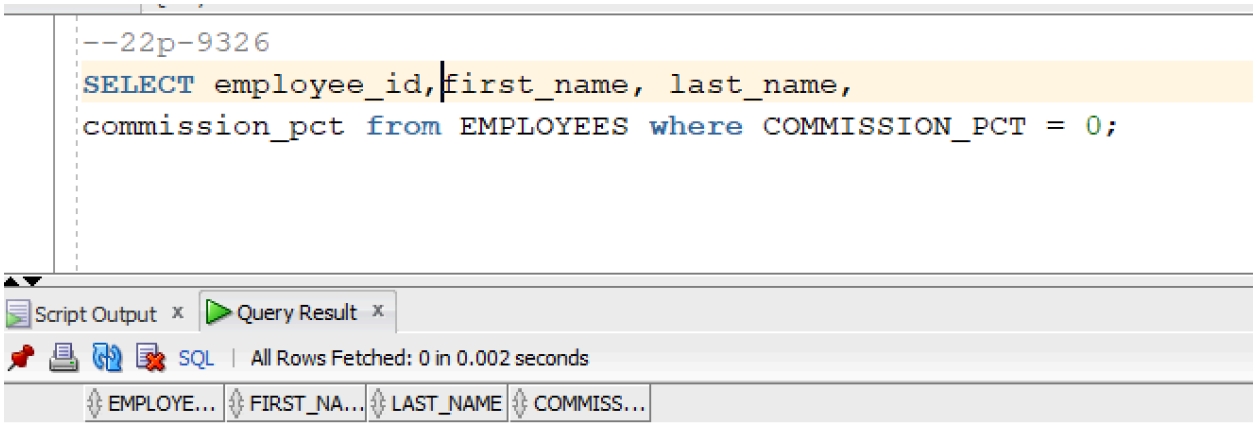
```
--22p-9326
SELECT employee_id, department_id, first_name from EMPLOYEES
where DEPARTMENT_ID BETWEEN 60 and 100;
```

Script Output | Query Result

SQL | All Rows Fetched: 49 in 0.006 seconds

	EMPLOYEE_ID	DEPARTMENT_ID	FIRST_NAME
1	100	90	Steven
2	101	90	Neena
3	102	90	Lex
4	103	60	Alexander
5	104	60	Bruce
6	105	60	David
7	106	60	Valli
8	107	60	Diana
9	108	100	Nancy
10	109	100	Daniel
11	110	100	John

3) List all the employees with no commission.



List all employees in order of their decreasing salaries.

```
--22p-9326
SELECT * from EMPLOYEES ORDER BY salary DESC;
```

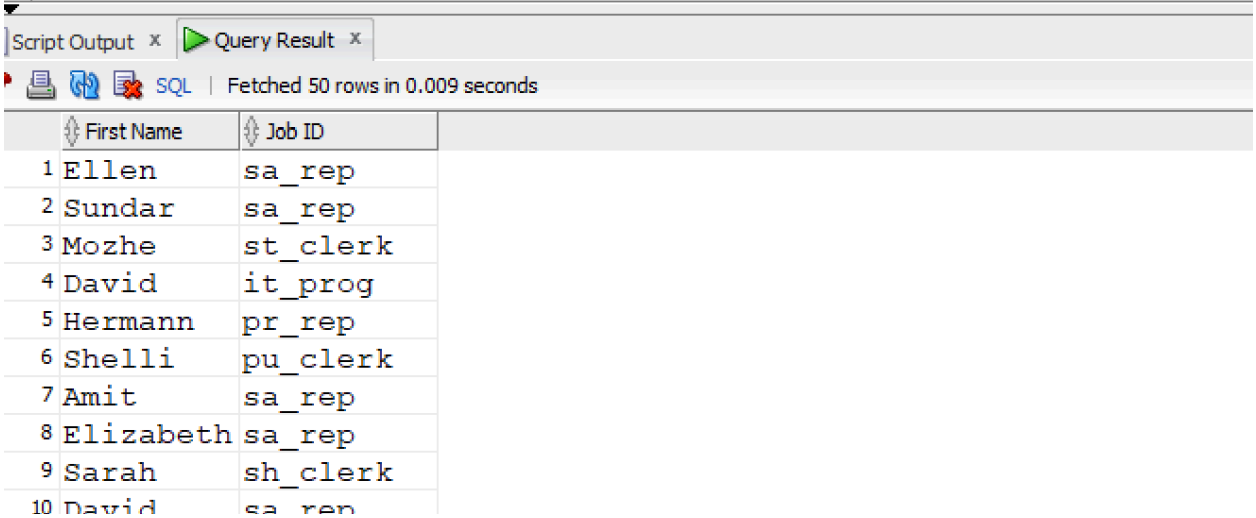
Script Output | Query Result

SQL | Fetched 50 rows in 0.012 seconds

		PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID
1		515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)
2	R	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100
3		515.123.4569	13-JAN-01	AD_VP	17000	(null)	100
4		011.44.1344.429268	01-OCT-04	SA_MAN	14000	0.4	100
5	R	011.44.1344.467268	05-JAN-05	SA_MAN	13500	0.3	100
6	E	515.123.5555	17-FEB-04	MK_MAN	13000	(null)	100
7	E	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101
8	S	515.123.8080	07-JUN-02	AC_MGR	12008	(null)	101
9	R	011.44.1344.429278	10-MAR-05	SA_MAN	12000	0.3	100
10		011.44.1343.929268	11-MAR-05	SA_REP	11500	0.25	148
11	L	515.127.4561	07-DEC-02	PU_MAN	11000	(null)	100
12	U	011.44.1344.619268	15-OCT-07	SA_MAN	11000	0.3	100
13		011.44.1644.429267	11-MAY-04	SA_REP	11000	0.3	149
14	Y	011.44.1344.429018	29-JAN-08	SA_MAN	10500	0.2	100
15	Y	011.44.1346.129268	11-NOV-05	SA_REP	10500	0.25	147
16		011.44.1345.429268	30-JAN-04	SA_REP	10000	0.35	146
17		011.44.1344.129268	30-JAN-05	SA_REP	10000	0.3	145
18		515.123.8888	07-JUN-02	PR_REP	10000	(null)	101

1) Print an employee name (first letter capital) and job_id(lower Case)

```
--22p-9326
SELECT INITCAP(first_name) as "First Name", LOWER(job_id)
as "Job ID" from EMPLOYEES;
```



Script Output | Query Result

SQL | Fetched 50 rows in 0.009 seconds

	First Name	Job ID
1	Ellen	sa_rep
2	Sundar	sa_rep
3	Mozhe	st_clerk
4	David	it_prog
5	Hermann	pr_rep
6	Shelli	pu_clerk
7	Amit	sa_rep
8	Elizabeth	sa_rep
9	Sarah	sh_clerk
10	David	sa_rep

2) For all employees employed for more than 100 months, display the employee number, hire date,number of months employed, first Friday after hire date and last day of the month hired.

```
--22p-9326
SELECT employee_id, hire_date,
FLOOR(MONTHS_BETWEEN(SYSDATE, hire_date)) AS "No Of Months",
NEXT_DAY(hire_date, 'MONDAY') AS "First Friday",
LAST_DAY(hire_date) AS "Last Date of Hired Month"
FROM employees
```

Script Output | Query Result

SQL | Fetched 50 rows in 0.006 seconds

	EMPLOYEE_ID	HIRE_DATE	No Of Months	First Friday	Last Date of Hired Month
1	100	17-JUN-03	259	23-JUN-03	30-JUN-03
2	101	21-SEP-05	232	26-SEP-05	30-SEP-05
3	102	13-JAN-01	288	15-JAN-01	31-JAN-01
4	103	03-JAN-06	228	09-JAN-06	31-JAN-06
5	104	21-MAY-07	212	28-MAY-07	31-MAY-07
6	105	25-JUN-05	235	27-JUN-05	30-JUN-05
7	106	05-FEB-06	227	06-FEB-06	28-FEB-06
8	107	07-FEB-07	215	12-FEB-07	28-FEB-07
9	108	17-AUG-02	269	19-AUG-02	31-AUG-02
10	109	16-AUG-02	269	19-AUG-02	31-AUG-02
11	110	28-SEP-05	232	03-OCT-05	30-SEP-05
12	111	30-SEP-05	232	03-OCT-05	30-SEP-05
13	112	07-MAR-06	226	13-MAR-06	31-MAR-06
14	113	07-DEC-07	205	10-DEC-07	31-DEC-07
15	114	07-DEC-02	265	09-DEC-02	31-DEC-02

3) Comparing the hire dates for all employees who started in 2003, display the employee number, hiredate, and month started using the conversion and date functions.

```
--22p-9326
SELECT employee_id,
hire_date,
TO_CHAR(hire_date, 'Month') AS "Starting Month"
FROM employees
WHERE EXTRACT(YEAR FROM hire_date) = 2003;
```

Script Output | Query Result | SQL | All Rows Fetched: 6 in 0.031 seconds

	EMPLOYEE_ID	HIRE_DATE	Starting Month
1	100	17-JUN-03	June
2	115	18-MAY-03	May
3	122	01-MAY-03	May
4	137	14-JUL-03	July
5	141	17-OCT-03	October
6	200	17-SEP-03	September

4) Find the next ‘Monday’ considering today’s date as date.

```
--22p-9326
SELECT NEXT_DAY(SYSDATE, 'MONDAY') as "Next Monday" from DUAL;
```

Script Output | Query Result | SQL | All Rows Fetched: 1 in 0.002 seconds

	Next Monday
1	03-FEB-25

5) List all Employees who have an ‘A’ in their last names.

```
--22p-9326
SELECT * from EMPLOYEES WHERE LAST_NAME LIKE '%A%';
```

Script Output | Query Result

SQL | All Rows Fetched: 4 in 0.003 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB
1	105	David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT_
2	130	Mozhe	Atkinson	MATKINSO	650.124.6234	30-OCT-05	ST_
3	166	Sundar	Ande	SANDE	011.44.1346.629268	24-MAR-08	SA_
4	174	Ellen	Abel	EABEL	011.44.1644.429267	11-MAY-04	SA_

6) Show all employees' last three letters of first name.

```
--22p-9326
SELECT SUBSTR(First_name, -3) as "LAst Three Letter"
from EMPLOYEES;
```

Script Output | Query Result

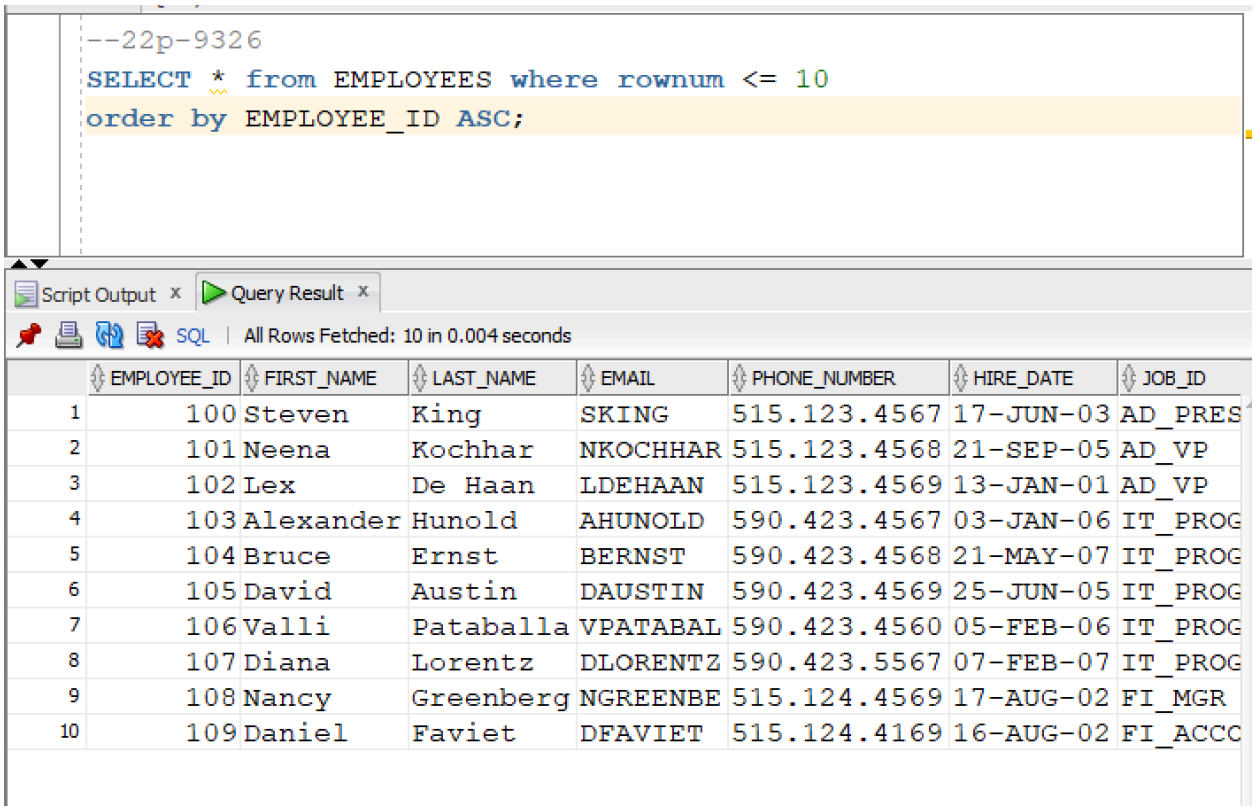
SQL | Fetched 50 rows in 0.006 seconds

	LAst Three Letter
1	len
2	dar
3	zhe
4	vid
5	ann
6	lli
7	mit
8	eth
9	rah

Home Task

1 – Fetch the bottom 10 values from the employees table w.r.t to employee

ID.



```
--22p-9326
SELECT * from EMPLOYEES where rownum <= 10
order by EMPLOYEE_ID ASC;
```

Script Output | Query Result

SQL | All Rows Fetched: 10 in 0.004 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP
4	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG
5	104	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG
6	105	David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT_PROG
7	106	Valli	Pataballa	VPATABAL	590.423.4560	05-FEB-06	IT_PROG
8	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-FEB-07	IT_PROG
9	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR
10	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCC

2 – List the departments with the most employees from top to bottom

(department ID can’t be NULL).

```sql
--22p-9326
SELECT department_id, COUNT(*) AS Total_Employee
FROM employees WHERE department_id IS NOT NULL
GROUP BY department_id
ORDER BY Total_Employee DESC;
```

Script Output | Query Result

SQL | All Rows Fetched: 11 in 0.004 seconds

	DEPARTMENT_ID	TOTAL_EMPLOYEE
1	50	45
2	80	34
3	100	6
4	30	6
5	60	5
6	90	3
7	110	2
8	20	2
9	70	1
10	10	1
11	40	1

3 - Display the employee number, name, salary of employee before and

after 15% increment. Name the calculated new salary as “Incremented Salary” and calculate the difference between two salaries. Name the increased amount to be “Incremented Amount”.

```
--22p-9326
SELECT employee_id, First_name || Last_name as "Full Name"
,salary,salary * 1.15 as "Increment Salary",(salary*1.15)-salary
as "Increment Amount" from employees;
```

Script Output | Query Result

SQL | Fetched 50 rows in 0.006 seconds

	EMPLOYEE_ID	Full Name	SALARY	Increment Salary	Increment Amount
1	100	StevenKing	24000	27600	3600
2	101	NeenaKochhar	17000	19550	2550
3	102	LexDe Haan	17000	19550	2550
4	103	AlexanderHunold	9000	10350	1350
5	104	BruceErnst	6000	6900	900
6	105	DavidAustin	4800	5520	720
7	106	ValliPataballa	4800	5520	720
8	107	DianaLorentz	4200	4830	630
9	108	NancyGreenberg	12008	13809.2	1801.2
10	109	DanielFaviet	9000	10350	1350
11	110	JohnChen	8200	9430	1230
12	111	IsmaelSciarra	7700	8855	1155

4 - Display the department and manager id wise avg commission for all

employees. Round the commission up to 2 decimals, filter any null.

values.

```sql
--22p-9326
SELECT ROUND(AVG(commission_pct), 2) AS Avg_Commission,
manager_id,
department_id
FROM employees
WHERE commission_pct IS NOT NULL
  AND department_id IS NOT NULL
GROUP BY manager_id, department_id;
```

Script Output x | Query Result x

SQL | All Rows Fetched: 6 in 0.005 seconds

	AVG_COMMISSION	MANAGER_ID	DEPARTMENT_ID
1	0.13	147	80
2	0.18	148	80
3	0.32	146	80
4	0.21	149	80
5	0.3	100	80
6	0.23	145	80